

25CY210 OPERATING SYSTEMS

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization & Digital logic and its design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility in acting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1

12L+0T+12P+12SL= 36 Hours

INTRODUCTION AND PROCESS MANAGEMENT:

Introduction: What Operating Systems do, Operating System Structure, Operating System Operations.
Process Concept: Process Concept, Process Scheduling, Interprocess Communication, Synchronization-The Critical-Section Problem, Peterson's Solution, Semaphores, Monitors

UNIT-2

MEMORY MANAGEMENT:

Basic concept, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table.

MASS STORAGE STRUCTURE:

File concept, Access Methods, Directory & Disk Structure, File-System Mounting, File Sharing, File System Structure, File System Implementation.

PRACTICES:

- Use the cat, sort, cut & paste commands on the data stored in file.
- Filters: cmp, comm, diff, head, tail, find, cut, paste, sort, uniq.
- Regular Expressions: grep, egrep, fgrep, Sed- line addressing, context addressing, text editing, substitution.

MODULE-2

UNIT-1

18L+0T+18P+18SL= 54 hours

PROCESS SCHEDULING:

Basic Concept, Scheduling Criteria, Scheduling Algorithms.

Deadlocks: Characterization, Methods for handling Deadlocks, Deadlock Prevention and Deadlock Avoidance.

UNIT-2

VIRTUAL MEMORY MANAGEMENT:

Demand Paging, Page Replacement: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU), Least Recently used (LRU), Allocation of Frames, Thrashing.

UNIT-3

STORAGE MANAGEMENT:

Implementing File Systems - Directory implementation, Allocation Methods, Free Space Management.

PRACTICES:

- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.
- Implementation of producer consumer problem.
- Consider the following page-replacement algorithms. Rank these algorithms on a five-point scale from “bad” to “perfect” according to their page-fault rate. Separate those algorithms that suffer from Belady’s anomaly from those that do not.
 - a) LRU replacement.
 - b) FIFO replacement.
 - c) Optimal replacement.
 - d) Second-chance replacement.
- Consider the page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.
- How many page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
 - a) LRU replacement
 - b) FIFO replacement
 - c) Optimal replacement
- How many page faults occur for your algorithm for the following reference string with four page frames? 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2.
- What is the minimum number of page faults for an optimal page replacement strategy for the reference string above with four page frames?
- Consider a demand-paged computer system where the degree of multiprogramming is currently fixed at four. The system was recently measured to determine utilization of the CPU and the paging disk. Three alternative results are shown below. For each case, what is happening?
 - a) Can the degree of multiprogramming be increased to increase the CPU utilization? Is the paging helping?
 - a) CPU utilization 13 percent; disk utilization 97 percent.
 - b) CPU utilization 87 percent; disk utilization 3 percent.
 - c) CPU utilization 13 percent; disk utilization 3 percent.
- Implementation of Disk scheduling algorithm–FCFS.
- Implementation of Disk scheduling algorithm–SSTF and SCAN.

ACTIVITIES:

- Developing the following solids with the help of cardboard/ thick paper, cube, cuboid prisms & pyramids (triangular, square, pentagonal, and hexagonal) right circular cylinder and cone.
- Design a Bolt and nut assembly for the given dimensions.
- Design a Connecting rod of IC engine for the given dimensions.
- Design a I.C Engine Piston with piston rings for the given dimensions.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Explain the concepts of operating System, including process scheduling and synchronization mechanisms.	Apply	1, 2
2	Analyse memory management techniques and files system structures to optimize storage and retrieval	Analyze	1, 2, 3, 5, 7
3	Apply process scheduling techniques and deadlock handling strategies to improve system efficiency.	Apply	1, 2, 7
4	Evaluate virtual memory management techniques, page replacement algorithms, and their impact on system performance .	Evaluate	3, 5
5	Design and Implement file system structures, disk scheduling algorithms, and free space management techniques for efficient storage utilization	Create	2, 5, 9

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):**TEXTBOOK:**

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 10th Edition, John Wiley & Sons, 2018.

REFERENCE BOOKS:

1. Sumitabha Das, Unix concepts and applications, TMH Publications, 4th Edition, July 2017.
2. Richard. Stevens and Stephen A Rago, "Advanced Programming in the Unix Environment", 3rd Edition, Addison-Wesley, 2013.
3. William Stallings, "Operating Systems-Internals and Design principles" PHI, 9th Edition, 2018.
4. Gary J. Nutt. Addison-Wesley, "Operating Systems: A Modern Perspective", 2nd Edition, 2001